RELATIVE PSEUDO-COHERENCE IS CHECKED ON AFFINE PAIRS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 16853-16887, immediately following the statement at 16836-16851.

Editorial note. One cross-chapter pointer (Schemes) plus four same-chapter statements are recorded in assembly.external_references.

Editorial note. The statement's condition on affine pairs is retained verbatim in the statement span.

Allowed standard prerequisites (editorial).

- The affine-pair criterion for one embedding. chapter lemma lemma-relatively-pseudo-coherent, cited in the statement and repeatedly in the proof. Supplies the equivalent conditions that the affine-local formulation refers to.
- Localisation of relative pseudo-coherence. chapter lemma lemma-localize-relative-pseudo-coherent, cited at proof lines 16880-16883. Moves the condition along the standard affine open U' inside the smaller pair.
- Glueing relative pseudo-coherence. chapter lemma lemma-glue-relative-pseudo-coherent, cited in the proof. Reassembles the local verifications into the global statement.
- Standard affine opens inside two affine opens. Schemes, lemma schemes-lemma-standard-open-two-affines, cited in the proof. Produces the standard affine open contained in the intersections appearing in the localisation argument.

Source reference. Original definition 09UI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16675–16675. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09VE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16789–16789. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09VD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16738–16738. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09VC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16610–16610. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IW: focused-seeds/geometry/S9-morph-rel-pseudo-coherence-characterize/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let E be an object of $D(\mathcal{O}_X)$. Fix $m \in \mathbf{Z}$. The following are equivalent*

1. *E is m-pseudo-coherent relative to S,*
2. *for every affine opens $U \subset X$ and $V \subset S$ with $f(U) \subset V$ the equivalent conditions of Lemma 09VC are satisfied for the pair $(U \to V, E|_U)$.*

Moreover, if this is true, then for every open subschemes $U \subset X$ and $V \subset S$ with $f(U) \subset V$ the restriction $E|_U$ is m-pseudo-coherent relative to V.

Proof. The final statement is clear from the equivalence of (1) and (2). It is also clear that (2) implies (1). Assume (1) holds. Let $S = \bigcup V_i$ and $f^{-1}(V_i) = \bigcup U_{ij}$ be affine open coverings as in Definition 09UI. Let $U \subset X$ and $V \subset S$ be as in (2). By Lemma 09VE it suffices to find a standard open covering $U = \bigcup U_k$ of U such that the equivalent conditions of Lemma 09VC are satisfied for the pairs $(U_k \to V, E|_{U_k})$. In other words, for every $u \in U$ it suffices to find a standard affine open $u \in U' \subset U$ such that the equivalent conditions of Lemma 09VC are satisfied for the pair $(U' \to V, E|_{U'})$. Pick i such that $f(u) \in V_i$ and then pick j such that $u \in U_{ij}$. By Schemes, Lemma 01IW we can find $v \in V' \subset V \cap V_i$ which is standard affine open in V' and V_i. Then $f^{-1}V' \cap U$, resp. $f^{-1}V' \cap U_{ij}$ are standard affine opens of U, resp. U_{ij}. Applying the lemma again we can find $u \in U' \subset f^{-1}V' \cap U \cap U_{ij}$ which is standard affine open in both $f^{-1}V' \cap U$ and $f^{-1}V' \cap U_{ij}$. Thus U' is also a standard affine open of U and U_{ij}. By Lemma 09VD the assumption that the equivalent conditions of Lemma 09VC are satisfied for the pair $(U_{ij} \to V_i, E|_{U_{ij}})$ implies that the equivalent conditions of Lemma 09VC are satisfied for the pair $(U' \to V, E|_{U'})$. □